2D MHD COMPUTER MODELING OF DENSE PLASMA FOCUS ACCELERATORS*

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Abstract[1]

Dense Plasma Focus (DPF) discharge is a very promising mechanism to achieve high neutron production yield (about 10^{11}–10^{12} per pulse) from the D-T reactions. Focus is conjectured to be a finite 2D Z-pinch formation near the end of the coaxial plasma accelerator, typically having the density above the level of $10^{19}/cm^3$ and temperature of a few keV during 100 to 150 ns. Interest to DPF appeared since the pioneer experiments of *Filippov* and the first theoretical reviews of *Dyachenko and Imshennik* and *Mather*. During the last decades theoretical investigation of DPF lagged behind the experiments, giving scant explanation of the experimental results. At the same time modeling and diagnostic capabilities have been dramatically improved. This presentation shows that the computer simulation by the code MHRDR (Magneto Hydro Radiative Dynamic Research) can help meet the goals and challenges of the LANL – Bechtel Nevada Dense Plasma Focus Accelerator project. Theoretical estimations, made in this report, represent the bounds between the parameters of source generator, geometry of the electrodes and feeding circuit and the initial density of the background gas in the form of the simple scaling laws.

I. INTRODUCTION

A plasma focus is an unusually dense high-temperature compact plasma formation, produced near the geometrical axis of system due to the cumulative effect. The action of the plasma focus is determined by the dynamic of the current sheath, which in its turn is affected by the geometry of the electrodes. In literature (see, for example, [1]) there exists a definition of two types of plasma focus, depending on the aspect ratio of the diameter of inner electrode (anode, in the most cases) D to its length l: $D/l > 1$ for Filippov [2] and $D/l < 1$ for Mather [3] types, respectively.

Also, there are two modes of plasma focus operation: low-energy, primarily, with the aim to obtain the powerful radiative source for EUV and x-ray ranges [3] , and high-energy neutron production [4] – [5] regimes. At high filling pressure range the thermal neutron-producing mechanisms are emphasized and become progressively dominant versus non-thermal, beam-target interaction, as the power level of the experiment is increased. This makes the magnetohydrodynamic (MHD) model even more applicable for simulation of dense plasma focus (DPF).

The numerical modeling of the dense plasma focus has been of interest since the pioneer works of *Potter* [6] and *D'yachenko* and *Imshennik* [7] . The dynamics of plasma focus has been analyzed in details in later publications, such as the work of *Maxon* and *Eddlman* [8] . Yet in none of the listed above papers the neutron production rate actually had been computed and correlated with the current waveform and the shock dynamic of the focus. Substantial progress on this field has been made in an extensive investigation of DPF, made by the *Lindemuth* and *Freeman* [4] . They used two-temperature MHD model, which started the history of MHRDR computer code (due to *Lindemuth* and *Sheehey*).

Advanced nowaday version of MHRDR code is based on the multi-temperature (ions, electrons and radiation) MHD model, which includes electron and ion thermal conduction, resistive diffusion and the radiation diffusion due to the final optical depth of plasma in addition to the Lorentz force and shock hydrodynamics. MHD model is solved self-consistently with an external circuit equation (which is capable of representing either capacitor bank or explosive magnetic compression generator, taking the data directly from actual experiments). Neutron yield is computed using Maxwell average cross sections for D-D reactions.

Basically, one can define three consecutive phases of plasma evolution in DPF:

1) breakdown of the background gas and formation of current sheath;
2) shock wave / current sheath acceleration;
3) shock wave / current sheath collapse at the axis of DPF and intense production of high-energy neutrons or EUV (x-ray) radiation.

MHRDR computer code is powerful and versatile tool, capable of modeling Plasma Focus Discharge at various regimes, such as low-density and high-density modes, as

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14. ABSTRACT

Dense Plasma Focus (DPF) discharge is a very promising mechanism to achieve high neutron production yield (about 10111012 per pulse) from the D-T reactions. Focus is conjectured to be a finite 2D Z-pinch formation near the end of the coaxial plasma accelerator, typically having the density above the level of 1019/cm3 and temperature of a few keV during 100 to 150 ns. Interest to DPF appeared since the pioneer experiments of Filippov and the first theoretical reviews of Dyachenko and Imshennik and Mather. During the last decades theoretical investigation of DPF lagged behind the experiments, giving scant explanation of the experimental results. At the same time modeling and diagnostic capabilities have been dramatically improved. This presentation shows that the computer simulation by the code MHRDR (Magneto Hydro Radiative Dynamic Research) can help meet the goals and challenges of the LANL Bechtel Nevada Dense Plasma Focus Accelerator project. Theoretical estimations, made in this report, represent the bounds between the parameters of source generator, geometry of the electrodes and feeding circuit and the initial density of the background gas in the form of the simple scaling laws.

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well as for the both types of electrode geometry. To have a better understanding of the governing processes at all phases of plasma evolution we have to consider in this presentation the specific DPF device. As an illustrative example we will take DPF at Los Alamos National Laboratory, usually referred as *Begay* plasma focus device. Basic parameters of this device are the following: radius of the inner electrode $r_{in} = 1.18\,\text{cm}$, radius of the outer electrode $r_{in} = 3.65\,\text{cm}$, and the length of the inner electrode $l = 15.7\,\text{cm}$ (Mather type configuration). Space between the electrodes is filled by deuterium with initial pressure of $1\,\text{Torr}$. Simplified scheme of the electric circuit, feeding the DPF device, is represented by the capacitor of $C = 36.4\,\mu\text{F}$, charge up to the voltage $U = 14\,\text{kV}$ and series inductance of $L = 178\,\text{nH}$. The equation for current dynamic is solved by the MHRDR code in self-consistent way together with the integration of MHD equations. The history of the basic parameters of electrical circuit is shown on Figure 1.

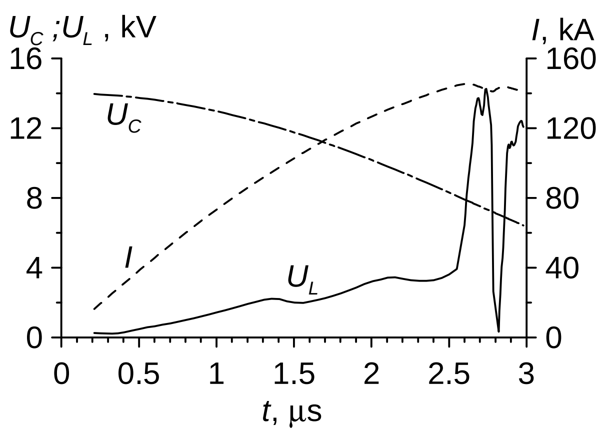

Figure 1. Source voltage U_C, current in the electrical circuit I, and the voltage drop between the electrodes of Plasma Focus U_L vs. time t.

II. DYNAMICS OF DPF DISCHARGE

A. Formation of current sheath

Generally, there are two types of the electrode-insulator layouts in DPF devices: schemes with horizontal and vertical placements of the insulator. Dynamic of the current sheath at the initial stage is shown on Figure 2 (horizontal insulator) and Figure 3 (vertical insulator). In both of these cases the initially ionized region is formed along the insulator, making the conducting bridge between the electrodes. Then the magnetic flux start to penetrate this slightly ionized plasma region from insulator, causing heating, ionization, and finally detaching of the hot dense current sheath from the surface of insulator. The only difference of these schemes is that in the case of horizontal placement of the insulator the current sheath is moving upward in the gap between the electrodes from the very beginning, yet in the case of vertical position of the insulator the “main” current sheath in plasma is formed in “indirect” manner, after the primary current sheath hits the outer electrode (see Figure 3). The inductance and plasma parameters in both cases are almost the same.

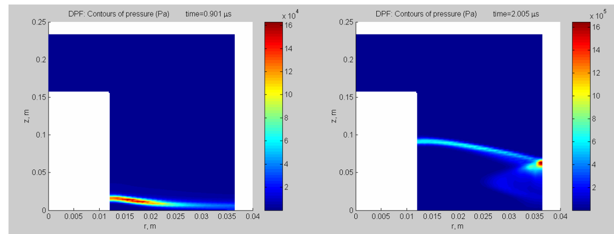

Figure 2. Pressure contours (in Pa) on the stage of the current sheath formation at times $t \approx 0.9\,\mu\text{s}$ and $t \approx 2\,\mu\text{s}$ at the case of horizontal placement of the insulator.

It should be noted that on the initial stage of current breakdown between electrodes some perturbations might arise along the insulator surface. This could affect negatively the stability of the current sheath. While in the case of “indirect” formation scheme the “main” current sheath is not affected by these undesired surface effects.

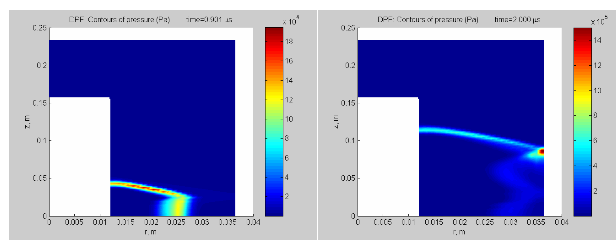

Figure 3. Sequence of pressure contours (Pa) on the stage of the current sheath formation at three moments: $t \approx 0.9\,\mu\text{s}$ and $t \approx 2\,\mu\text{s}$ at the case of vertical placement of the insulator, positioned at the lower end of inner electrode.

B. Acceleration of the current sheath

Once the slim plasma region is created near the insulator surface, it is subject to the action of Lorentz force. The magnetic field pressure increases with the current rising (see Figure 1), and, since there is no any counteraction from the plasma region, the current sheath detaches from the insulator surface and starts to accelerate upward.

The slices of plasma parameters through the moving current sheath at different moments of time are shown on Figure 4. As we can see no substantial amount of plasma density or magnetic flux is left behind the accelerating current sheath. The current sheath itself has complex structure. In the dense and relatively cold front part the dominant effect is plasma accretion due to snowplow scenario, and heating there occurs due to the hydrodynamic shock compression. The hydrodynamic shock wave is followed by the "magnetic piston" that feeds the shock with the kinetic energy resulting from the magnetic field pressure.

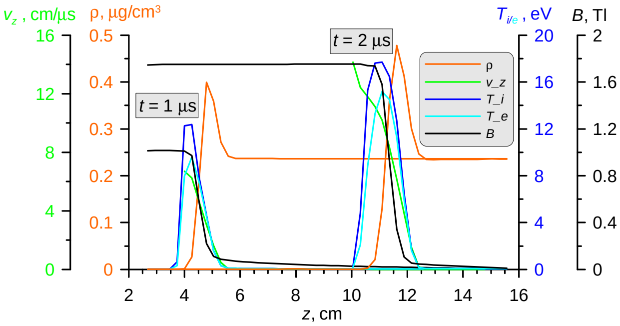

Figure 4. Parameters of moving current sheath (plasma density r , temperatures of ion T_i and electron T_e components, intensity of the magnetic field B , and axial component of the velocity v_z) at two consecutive moments of time $t = 1\ ms$ and $t = 2\ ms$. Dynamics of current sheath is due to the snowplow scenario: the hydrodynamic shock is followed by the "magnetic piston". The slice of plasma parameters is taken along z -axis at $r = 1.45\ \text{cm}$ (or at $0.27\ \text{cm}$ from the surface of inner electrode).

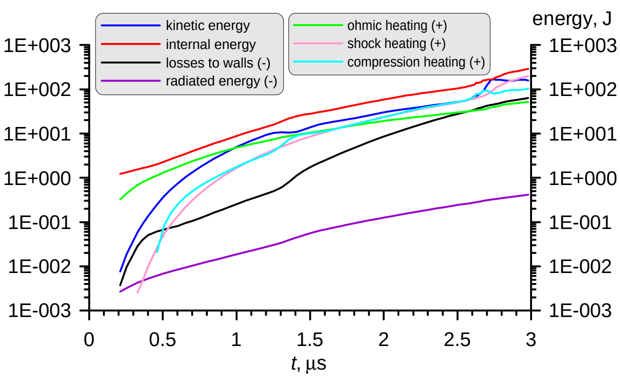

Figure 5. Energy balance during the evolution of plasma focus discharge.

The role of the ohmic heating is dominant at the stage of the current sheath formation. But later, during the acceleration phase, the temperature of the current sheath is defined mainly by the shock compression, as it can be seen from Figure 5. Thus the accelerating plasma bypasses highly dissipative regime that is so typical for plasmas at diffuse z-pinches. This explains extremely low resistance of the current sheath (see Figure 1: the value of current in the circuit is about $\sim 50 - 100\ \text{kA}$, while the voltage drop between the electrodes of plasma focus is $\sim 1 - 2\ \text{kV}$) and the fact that the electron temperature is lower than the ion one during all phases of DPF evolution.

The radiated energy is negligible, compared with the shock and ohmic heating, and thus does not affects much the plasma dynamics. But *Lindemuth* and *Freeman* reported in Ref. [4] that for more dense and energetic plasma focus (filling pressure $\approx 22\ \text{Torr}$, maximum current of $\approx 3\ \text{MA}$) the radiation is able to change substantially the energy balance of DPF discharge.

C. Formation of z-pinch and neutron production

After the collapse of the accelerated current sheath the high-energy, high-density z-pinch discharge region is formed at the geometrical axis of DPF. This substantially non-uniform process is governed by 2D-effects. As it can be seen from the pressure contour plots on Figure 6, the collapse starts near the surface of inner electrode and then propagates upward along z -axis, like in the case of the conical z-pinch discharge.

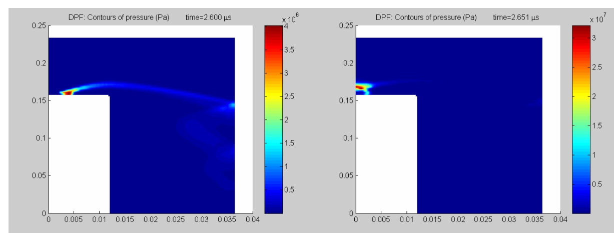

Figure 6. Final stage of plasma focus discharge evolution: collapse of the current sheath at the geometrical axis of the device. Plasma pressure contours are shown at two consecutive moments of time: $t \approx 2.6\ \mu\text{s}$ and $t \approx 2.65\ \mu\text{s}$.

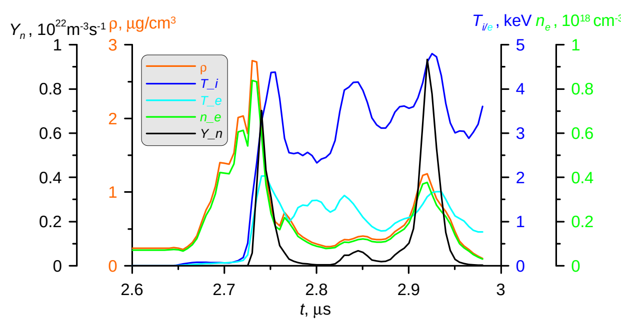

Figure 7. Evolution of plasma parameters in z-pinch discharge at the geometrical axis of plasma focus. The figure shows plasma density r , temperatures of ion T_i and electron T_e components, electron density n_e , and local neutron production rate Y_n versus the time t at the axis point $z \approx 18.1\ \text{cm}$, which is located $\approx 2.5\ \text{cm}$ above the inner electrode surface.

The local neutron production rate $Y_n(r, T_i)$ correlates with plasma density and ion temperature. Two peaks at the history of this function at the point $(r = 0\,;\, z = 18.1\ \text{cm})$ that is $2.5\ \text{cm}$ above the surface of the inner electrode can be observed on Figure 7 at the moments $t = 2.74\ \mu s$ and $t = 2.92\ \mu s$. The former peak is due to the high density, while the latter one is due to the high ion temperature. The nature of these peaks can be explained by Figure 8 that gives us radial distribution of plasma parameters at the level $z = 18.1\ cm$ at the moments $20\ ns$ before these peaks occur.

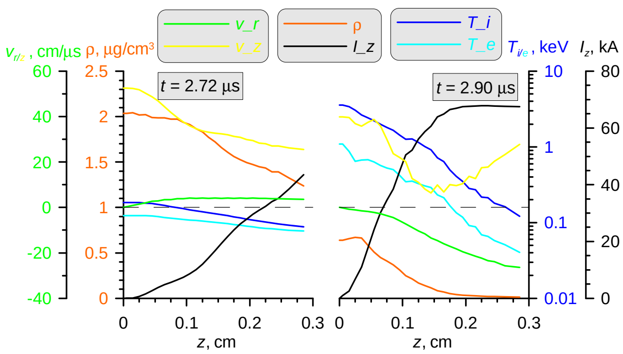

Figure 8. Radial distribution of plasma parameters (plasma density r, temperatures of ion T_i and electron T_e components, radial v_r and axial v_z components of the velocity, and total enclosed current through plasma I_z, vs. the radial coordinate r) at the center of the discharge. Data slice is taken at level $z = 18.1\ cm$ at two consecutive moments of time $t = 2.72\ \mu s$ (on the left) and $t = 2.9\ \mu s$ (on the right). Black dashed lines represent zero level for the values of both velocity components.

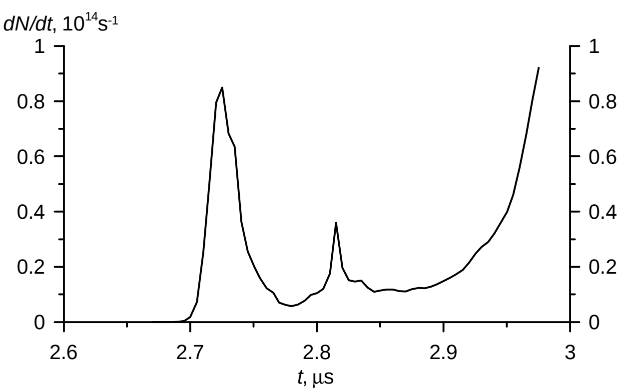

Figure 9. Total (space integrated) rate of neutron production $dN/dt, 10^{14}\ s^{-1}$ versus time.

As it was mentioned above, the collapse of the accelerated plasma layers starts near the surface of inner electrode. High-energy, high-density plasma is formed at the region, which is confined by the high magnetic field pressure in radial direction and the electrode surface itself in axial direction. The only way that plasma can escape is to move in upward direction, along z-axis. Figure 8 says that at $t = 2.72\ \mu s$ the magnetic piston is still far from z-axis (at level $z = 18.1\ cm$). But due to the strong upstream flow of hot dense plasma along z-axis ($v_r << v_z$), the axial density increases dramatically giving the first peak of the local neutron production rate. At the moment $t = 2.90\ \mu s$ the magnetic piston is just about to arrive at z-axis, compressing in radial direction all plasma layers in front of it. The second peak of neutron production rate, due to the hydro-magnetic shock collapse at the axis, is higher than the first one, having pure hydrodynamic nature.

The evolution of space-integrated rate of neutron production is presented on Figure 9.

III. CONCLUSION

The results of the computer modeling of PDF device give the adequate physical scenario of the evolution of plasma focus discharge. Simulations, made by the MHRDR code (including that in Ref. [4]), are in good qualitative and quantitative agreement with the data from actual experiments (LANL, etc.). MHRDR is versatile and “ready-to-use” tool for modeling the variety of DPF discharges.

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